

Stanley Zhou

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Seeking May–August 2027 AI/ML Internships

EDUCATION

University of Waterloo

Waterloo, Ontario

Honours Bachelor of Mathematics in Computer Science (Co-op)

September 2026 – May 2031

- Awards: President's Scholarship of Distinction (\$2,000), Microsoft Alumni Entrance Scholarship (\$2,000)
- Relevant Coursework: Linear Algebra, Proof-based Math, Calculus, Data Structures and Algorithms, Software Engineering, Statistical Analysis

EXPERIENCE

Computer Vision & Machine Learning Researcher

September 2026 – Present

Weizmann Institute of Science – Faculty of Physics, [Ant Collective Behavior Group](#) Rehovot, Israel (Remote)

- Designed computer vision pipelines using SLEAP for a **52-camera 4K apparatus**, detecting, identifying, and tracking **700–1,000 ants** in 3D space at **10 fps**.
- Analyzed ant poses and postures using head, thorax, and abdomen landmarks; tracked ant scales and collective chains functioning as weights or zipper-like structures.
- Developing multi-view geometry, cross-camera association, and motion modeling techniques to handle dense ant interactions and partial occlusions.
- Contributing to current experiments and research on individual and collective ant behavior.

Economic Data Research Assistant

December 2023 – September 2024

Smith School of Business – Queen's University, mentored by Dr. Bhargav Gopal

Kingston, Ontario

- Contributed to a forthcoming corporate governance paper on firms' operating performance following the repeal of government-mandated leadership quotas (California SB-826).
- Analyzed **12 papers** on corporate governance, shareholder rights, valuation, and performance indicators; presented findings at internal meetings to inform the literature review.
- Fit Fama–French 3-Factor and Carhart 4-Factor models in R on **over 6 million observations**.

PROJECTS

VLA Trajectory Data Optimization Research Project

Python, PyTorch, NumPy, scikit-learn, Jupyter, AWS Cloud

- Reviewed Vision-Language-Action (VLA) literature on curating and synthesizing rare robot trajectories.
- Created a pipeline to segment trajectories into primitives and distill smaller, distribution-matched training sets; applied oversampling to cluster trajectories and address underrepresentation.

[Wat Way](#) – Hackathon Project for General Catalyst Hackathon

A Algorithm, Max-Flow, TypeScript, React*

- Created a campus map and routing tool using an A* algorithm to efficiently navigate between rooms, buildings, and tunnels at the University of Waterloo.
- Accounted for weather exposure, foot traffic, and floor plans with max-flow analysis to identify routing bottlenecks.

[BananaMOV](#) – Hackathon Project for JamHacks 10

Next.js, TypeScript, Gemini API, ElevenLabs API | Won “Best use of ElevenLabs” prize track

- Built a full-stack app that scores uploaded videos using Gemini 2.5 Flash for visual composition plans and ElevenLabs' Music API for original, timeline-matched music.
- Used Clerk for authentication, Neon Postgres for artifact storage, and Demucs for audio stem separation.

TECHNICAL SKILLS

Languages: Python, R, Java, JavaScript/TypeScript, HTML/CSS

Frameworks & Libraries: PyTorch, NumPy, Pandas, Matplotlib, dplyr, React, Node.js, Django

Developer Tools: Git, Linux/Unix, CLI/Bash, Google Cloud Platform